
Richard Waters

Senior Software Engineer | Secure Systems | Cybersecurity | Cloud Platforms

[Email](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL PROFILE

Senior software engineer with over 10 years of programming experience, delivering secure, scalable software across backend platforms, cybersecurity tooling, cloud services and AI-enabled products. Strong engineering fundamentals support confident movement between languages, frameworks and established codebases while respecting each technology's conventions, architecture and ecosystem. Skilled at learning how an organisation operates, identifying what its customers and users genuinely need, and translating incomplete or complex requirements into clear, maintainable software. Experienced across API design and integration, data-intensive workflows, CI/CD, containerised delivery, secure systems design, testing, technical mentoring and collaboration with both technical and non-technical stakeholders. Comfortable taking ownership from early discovery and prototyping through implementation, deployment, monitoring and continued improvement.

“A language that doesn't affect the way you think about programming is not worth knowing.”

— Alan Perlis

TECHNICAL SKILLS

Programming Languages & Codebases

- Python, TypeScript/JavaScript, Go, Rust, C++, Bash and PHP
- Confident across greenfield products, legacy applications, monorepos, microservices and large existing codebases

Backend & Frameworks

- Node.js, Express, Flask, FastAPI, Spring Boot, Codelgniter and server-side Next.js
- API architecture, asynchronous processing, authentication, background jobs and third-party integrations

Frontend & Product UI

- Next.js, React, Vue, Shopify/Liquid, Tailwind CSS and component-based design systems
- Responsive interfaces, accessibility, performance optimisation and customer-facing workflows

Cloud & Infrastructure

- AWS and Microsoft Azure; scalable architecture, high availability and cost optimisation
- Infrastructure as Code with Terraform, CloudFormation and ARM templates

DevOps, Containers & Delivery

- GitHub Actions, GitLab CI, Jenkins, Azure DevOps, Docker and Kubernetes
- Automated build, test and deployment pipelines with monitoring, observability and rollback awareness

AI & Machine Learning

- PyTorch, TensorFlow, scikit-learn, computer vision, NLP and production data pipelines
- Model evaluation, explainability, MLOps workflows and human-centred AI integration

Cybersecurity & DevSecOps

- Application, network and cloud security; threat modelling and vulnerability assessment
- OAuth2, OpenID Connect, RBAC, JWT, OWASP ZAP, Burp Suite and Snyk
- Linux hardening, permissions, iptables, SSH, fail2ban, ISO 27001 and GDPR

Data Engineering & Databases

- PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Neo4j and Redis
- Data modelling, ETL workflows, caching, query optimisation and streaming with Apache Kafka

Testing & Quality Assurance

- Unit, integration, end-to-end, regression, security and performance testing
- pytest, Jest, Selenium, Cypress, TDD, mocking, test coverage and CI quality gates

Architecture & Engineering Practice

- REST and GraphQL APIs, event-driven systems, distributed services and design patterns
- Code review, refactoring, technical debt reduction, documentation and maintainability

Mobile, Robotics & Automation

- React Native, Flutter, ROS, sensor integration, control systems and automation workflows

PROFESSIONAL EXPERIENCE

Senior Software Developer

Club L London | Manchester | 2022 - 2024

- Engineered scalable Node.js backend services, asynchronous processing pipelines and secure API integrations for high-volume retail operations.
- Introduced CI/CD automation with GitHub Actions and Docker, improving release consistency, deployment confidence and operational resilience across shared environments.

Cybersecurity Engineer

ShadowFox Security | Remote | 2020 - 2022

- Designed security tooling, assessment dashboards and Python automation for repeatable penetration testing and vulnerability workflows.
- Led red-team-style exercises, built a React and Flask OWASP training platform, and translated technical findings into clear remediation priorities for technical and non-technical stakeholders.

Full Stack Developer

Liverpool Hope University | Liverpool | 2020 - 2021

- Delivered a PHP and CodeIgniter CRM supporting grants and administrative workflows for internal university teams.
- Translated non-technical requirements into submission, tracking and feedback workflows that improved operational handling.

Frontend Developer

Velstar | Liverpool | 2018 - 2020

- Delivered Shopify themes and customer-facing applications for major retail brands, working closely with design and marketing teams on reliable releases.

KEY PROJECTS

LukaScope

[GitHub](#) • [Demo](#)

AI-assisted medical imaging platform for blood-smear analysis and diagnostic review. Combines a modern Next.js interface with secure backend services, a Python inference workflow, result history and explainability imagery to make model output clearer for users.

DockerScripts

[GitHub](#)

Reusable container environments that standardise local development, testing and deployment across different technology stacks. Designed to reduce setup friction, keep services reproducible and make onboarding into unfamiliar codebases more consistent.

Password Cracker

[GitHub](#)

Interactive cybersecurity education tool demonstrating dictionary and brute-force password-cracking techniques. Presents attack behaviour clearly so users can understand password strength, search-space growth and the practical value of stronger authentication.

MalViz

[GitHub](#)

Local-first malware-analysis MVP for quarantining suspicious files, performing static checks and producing readable scan reports. Designed around safe handling, transparent findings and turning low-level security information into practical guidance.

QUALIFICATIONS

BSc (Hons) Computer Science • First Class Honours • Liverpool Hope University